Chapter 11

Agricultural Microbiology

Chapter Outline

11.1 Biogeochemical Cycles

11.2 Biofertilizers

11.3 Biopesticides

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The *Anabaena azollae-Azolla* association is a symbiotic relationship between a blue green alga and a rice plant. *Azolla* is a small, fast growing floating water fern. In the cavities of leaves of azolla, *Anabaena azollae* a blue green algae (a cyanobacterium) fixes nitrogen from the air.

Learning Objectives

After studying this chapter the student will be able,

- *To understand the various biogeo chemical cycle*
- *To know the nitrogen fixation process*
- *To infer about the biofertilizer*
- *To learn the role of biopesticides in agriculture*

Agricultural microbiology is a branch of microbiology which deals with the study of microorganisms that are involved in various agricultural processes taking place in soil, plants and agro industries.

Agriculturally important processes like the Biological nitrogen fixation, nutrient cycling of Carbon, Sulphur, Phosphorus and Nitrogen are microbe mediated and play a very significant role in replenishing the nutrient supply in soil. Certain microorganisms like bacteria, fungi and viruses are economically important since they cause plant diseases and are responsible for severe crop losses. The study that deals with plant diseases is called plant pathology.

There are microorganisms capable of producing plant hormones thereby enhancing the plant growth. Some micro organisms can be used as biopesticides as they have ability to kill insects that damage plants.

11.1 Biogeochemical Cycles

Biogeochemical cycling is defined as the movement and conversions of materials by biochemical activities throughout air, water and soil. All living organisms participate in biogeochemical cycling but